

Avni Girish

(732) 829-9689 | avni123.girish@gmail.com | [linkedin.com/in/avni-girish](https://www.linkedin.com/in/avni-girish) | github.com/avnigirish | www.avnigirish.com | US Citizen

EDUCATION

Rutgers University–New Brunswick

New Brunswick, NJ

Bachelor of Science in Computer Science and Cognitive Science - Double Major

May 2026

- **Relevant Coursework:** Intro to Artificial Intelligence, Data Science, Data with R, Algorithms, Data Structures, Discrete Structures I & II, Software Methodology, Information and Data Management, Computer Architecture

WORK & LEADERSHIP EXPERIENCE

Data Science & AI Engineer Intern

Lake Mary, FL

BNY Mellon

Jun 2025 - Aug 2025

- Built multi-LLM agent framework (Claude 4, GPT-4.1) to automate UI generation, reducing development cycles from 72 hours to 5 minutes
- Developed AI-driven Chat interface enabling natural language UI generation, reducing manual revisions by 65% and increasing analyst throughput 3.5x
- Produced LangChain-based agent framework with FastAPI backend enabling autonomous task routing between planning and coding LLMs and Advanced Prompt Engineering reducing manual handoffs 40%

IT Business Analyst Intern

Trenton, NJ

New Jersey Economic Development Authority

Jun 2024 - Aug 2024

- Translated business requirements for 5+ enterprise projects (\$500K+ budget) into technical user stories, enabling 10-person dev team to deliver 100% on schedule using **JIRA, Dynamics365, and Visio**
- Facilitated release planning for 20+ milestones across cross-functional teams, proactively removing roadblocks to maintain 100% on-time delivery record
- Analyzed competitive landscape across 5 states, providing data-driven recommendations that shaped policy decisions impacting 250+ business applicants

Undergraduate Research Assistant

New Brunswick, NJ

Cognition and Learning Center

Jan 2024 - Present

- Engineered data collection pipelines in **Python/MATLAB** capturing 10K+ gameplay data points for ML feature engineering / prediction
- Analyzed 3GB+ of gameplay data using Python, identifying 25% recall improvement patterns now used to optimize educational content recommendation logic
- Applied sentiment analysis to 200+ surveys, surfacing motivational drivers for AI engagement features

Rutgers Computing Academic Tutor

New Brunswick, NJ

Rutgers University Department of Computer Science

Jan 2024 - Present

- Tutored 650+ students in Data Structures/Algorithms using **Java**, driving 15% course pass rate improvement
- Led bi-weekly faculty-student sync meetings analyzing 200+ survey responses, resulting in curriculum updates and increased student satisfaction scores by 25%

PROJECTS

Food Project Research Lead

New Brunswick, NJ

Cognition and Learning Center

May 2025 - Present

- Constructed a comprehensive R dataset from 200+ participants combining cognitive and microbiome data for downstream ML modeling
- Supervised 4+ RAs; automated validation in R cut processing time 40%

[AI-Powered Personal Portfolio & Interactive Resume](#)

Aug 2025 - Present

- Engineered a full-stack portfolio with Admin backend using **Next.js 15, TypeScript, Tailwind CSS, OpenAI API, GIT and REST APIs** and hosted on Vercel
- Devised a multi-LLM orchestration layer with OpenAI API and OpenRouter providing intelligent routing and fallback across models

MoodSync+: A comprehensive Mental and Physical Wellness App

Dec 2024 - Present

- Developed **FastAPI/Streamlit/PostgreSQL** platform integrating Spotify API to deliver 50+ personalized mood-based recommendations
- Implemented NLP-based sentiment analysis with **Hugging Face and OpenAI APIs**, achieving 95% accuracy in emotion detection for journal entries
- Created interactive dashboards with **Matplotlib and Plotly** to visualize mood trends for 100+ users, backed by a **PostgreSQL** database

SKILLS, ACTIVITIES & INTERESTS

Programming Languages: Python, Java, C, C#, C++, R, JavaScript, SQL

AI/Machine Learning Tools: LangChain, Hugging Face, PyTorch, TensorFlow, OpenAI/Anthropic APIs, RAG Systems, Vector Databases, NLP, Sentiment Analysis, EDA, Pandas, Scikit-learn, Pydantic, Prompt Engineering

Frameworks: FastAPI, React.js, Next.js, Flask, Node.js, Streamlit

Databases: MySQL, PostgreSQL, MongoDB, SQLite, Firebase

Developer Tools: Git, JIRA, MATLAB, Unity, VS Code, Github CoPilot, KiloCode

Certifications & Training: Python Programming - *GirlsComputingLeague* (2020), Software Engineering Fellowship - *Rutgers Blueprint* (2023), CS50: Introduction to Computer Science - *Harvard University* (2023), Social/Behavioral/Epidemiologic Research Investigators - *CITI Program* (2023), *Amazon University Event: AWS Reachback Cohort* (February 2026 - April 2026), AWS Cloud & AI Practitioner Certificate (May 2026)

Awards: 1st place in the Advanced Category: Software Engineering/Software Developer - *HackSBPL Hackathon* (2021)

Organizations: Undergraduate Alliance of Computer Science, Women in Computer Science, Rutgers University Programming Association, Women in Science and Engineering Program-Douglass Residential College